

STUDY PROTOCOL

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Senolytics To sLOw Progression of Sepsis (STOP-Sepsis) in elderly patients: Study protocol for a multicenter, randomized, adaptive allocation clinical trial

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Abstract

Background Senescent immune cells exhibit altered gene expression and resistance to apoptosis. The prevalence of these cells increases with age and emerging data implicate senescence-associated maladaptive signaling as a potential contributor to sepsis and septic shock. The senolytic drug fisetin promotes clearance of senescent cells and is hypothesized to mitigate septic responses to infection.

Methods We are conducting a multi-center, randomized, double-blinded, adaptive allocation phase 2 clinical trial to assess the efficacy of the senolytic drug fisetin in preventing clinical deterioration of elderly patients diagnosed with sepsis. We intend to enroll and randomize 220 elderly patients (age > 65) with the clinical diagnosis of sepsis to receive either fisetin as a single oral dose of 20 mg/kg, fisetin in two oral doses of 20 mg/kg each spaced 1 day apart, or placebo. The primary outcome will be changed in the composite of cardiovascular, respiratory, and renal sequential organ failure assessment scores at 7 days from enrollment. Secondary outcomes include quantification of senescent CD3+ cells at 7 days, and 28-day assessments of organ failure-free days, days in an intensive care unit, and all-cause mortality.

Discussion This multi-center, randomized, double-blinded trial will assess the efficacy of fisetin in preventing clinical deterioration in elderly patients with sepsis and measure the effects of this drug on the prevalence of senescent immune cells. We intend that the results of this phase 2 trial will inform the design of a larger phase 3 study.

Trial registration This trial is registered to ClinicalTrials.gov under identifier NCT05758246, first posted on March 7, 2023.

Keywords Sepsis, Septic shock, Senescence, Cellular senescence, Senolytic, Fisetin, Aging

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Administrative information

Note: the numbers in curly brackets in this protocol refer to SPIRIT checklist item numbers. The order of the items has been modified to group similar items (see <http://www.equator-network.org/reporting-guidelines/spirit-2013-statement-defining-standard-protocol-items-for-clinical-trials/>).

Title {1}	Senolytics To sLOw Progression of Sepsis (STOP-Sepsis) in elderly patients: study protocol for a multicenter, randomized, adaptive allocation clinical trial
Trial registration {2a and 2b}	This trial is registered to clinicaltrials.gov under identifier NCT05758246, first posted on March 7, 2023.
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Role of sponsor {5c}	Dr. Puskarich is the investigator-sponsor for the study. He was directly involved in trial design, and will be involved with subject recruitment, data collection and interpretation, and manuscript preparation. He has ultimate oversight authority over all trial activities at all sites. The administrative and funding instrument used for this study is the cooperative agreement, an "assistance" mechanism in which substantial National Institute on Aging (NIA) programmatic involvement with the investigators is anticipated during the performance of the activities based on the terms and condition of the U01AG076929 award. The NIA purpose is to support and stimulate the recipients' activities by involvement in and otherwise working jointly with the recipients in a partnership role; it is not to assume direction, prime responsibility, or a dominant role in the activities. The NIA must approve the study protocol and protocol changes.
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Introduction
Background and rationale {6a}

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection [1]. Clinically, this is manifested when a patient with a known or suspected infection has an increase in sequential organ failure assessment (SOFA) score of 2 points or more [2], and septic shock occurs when a patient with sepsis develops hemodynamic instability and elevated blood lactate levels [3]. Despite campaigns to promote early recognition and treatment, these diagnoses continue to cause significant mortality worldwide [4] and represent a significant financial burden on the healthcare system accounting for over \$20 billion in healthcare costs annually in the USA alone [5].

The underlying biomechanisms of sepsis are driven at least in part by a maladaptive immune response by which self-injurious inflammation leads to multiple organ dysfunction syndrome (MODS) [1, 6]. A complex, homeostatic anti-inflammatory response often follows [6], ultimately leading to broad dysfunction across both the adaptive and innate immune systems, including abnormal cytokine signaling, release of premature neutrophils, and impairment of cytotoxic functions and antibody production [7–9].

Both the incidence and mortality of sepsis increase with age [10–12]. This is consistent with preclinical sepsis models of aged rodents that demonstrate higher mortality and inflammation [12]. Additionally, older patients often manifest protracted organ failure, a lower likelihood of discharge to home, worse performance of activities of daily living (Zubrod performance), and higher long-term mortality [13].

Senescence represents an alternative cellular fate from apoptosis or necrosis, characterized by altered gene and

protein expression with resistance to apoptosis [14, 15]. Senescent cells (SnCs) are increasingly prevalent with age [16, 17], and can be found in organs throughout the body. SnCs can be quantified by tissue biopsy or circulating peripheral blood mononuclear cells (PBMCs), particularly CD3+ cells [17]. Tissue-based SnCs may contribute to organ dysfunction, while age-related immunosenescence is characterized by broad disruptions in innate, T, and B cell function [9, 18, 19]. These changes closely mimic changes observed in sepsis. It remains unclear if these similarities represent coincidentally overlapping mechanisms or age-related changes that predispose to increased sepsis incidence and severity. Additionally, the secretory proteins observed from SnCs in vitro overlap significantly with known sepsis mediators such as cytokines and chemokines, and exposure to lipopolysaccharide (LPS), an antigen known to provoke a septic response, resulted in both increased SnC formation, and increased secretory output from existing SnCs. While the role of the dysfunctional immune response in sepsis is well established, and multiple groups have observed immunosenescence during sepsis [7, 9, 18], it remains unclear if manipulation of senescent cells affects sepsis outcomes. Due to the association of senescent cell formation with aging and a number of disease conditions, a recent National Institute of Aging (NIA) workshop recommended the study of drugs targeting senescence (senolytics) in phase II trials [20].

Fisetin is a flavonol with senolytic properties and is present in fruits and vegetables. In animal models of sepsis, it reduces MAPK/NF- κ B-induced MODS, improving endothelial [21], lung [22], and kidney function [23], and reducing mortality [21]. Additionally, fisetin has a favorable safety profile, low incidence of reported side effects, and its effectiveness as a senolytic does not require a sustained plasma level [24, 25]. This allows for bolus dosing and circumvents the need for detailed pharmacodynamic considerations that typically challenge the translation of therapeutics to humans. We have now initiated a phase 2 multi-center, randomized, controlled, double-blinded study of fisetin as a therapy to treat sepsis.

Objectives {7}

Our broad hypothesis is that bolus dose fisetin early in the course of a septic response will immediately and durably reduce the prevalence of senescent cells in a dose-dependent manner, reduce systemic inflammation, and demonstrate preliminary efficacy in reducing multi-organ dysfunction during sepsis. Our specific biological objectives include assessment of peripheral CD3+ senescent immune cells at 7 and 28 days and assessment of markers of systemic inflammation at 7 days. Our specific clinical objectives include assessment of the change in

composite cardiovascular, respiratory, and renal SOFA score at 7 days, change in total SOFA score at 7 days, Zubrod performance score and SF-12 score at 7 and 28 days, days in ICU at 28 days, organ failure-free days at 28 days, all-cause mortality at 28 days, and occurrence of safety events associated with fisetin use.

Trial design {8}

This trial is designed as a multi-center, randomized, double-blinded, parallel-group, response-adaptive superiority trial with a planned sample size of 220 participants. Subjects will initially be randomized to receive (a) oral fisetin as a single 20 mg/kg oral dose, (b) oral fisetin as two 20 mg/kg oral doses spaced 1 day apart, or (c) placebo at a 1:1:1 ratio. After enrollment of the first 60 subjects and at every 20 patients thereafter, the randomization ratios will change. While the proportion of subjects allocated to placebo will remain one in three to mitigate the effect of temporal changes in standard of care, the allocation ratios for the two treatment arms will be adapted based on the 7-day change from baseline in the composite cardiovascular, respiratory, and renal sequential organ failure assessment score (Δ CRR-SOFA score) using bayesian response adaptive randomization, with the goal of allocating more patients to the arm that appears more efficacious at the time of the most recent randomization. Further details on this will be provided within the “Sequence generation {16a}” section. Randomization will not be stratified.

Methods: participants, interventions, and outcomes

Study setting {9}

Participants were initially enrolled from one tertiary academic medical center, one tertiary referral county hospital, and two community hospitals. As of February 2024, we have expanded to include a third community hospital (Table 1).

Eligibility criteria {10}

Inclusion criteria

1. Age ≥ 65 years
2. Primary diagnosis of acute infection (per investigator judgment)
3. SOFA score ≥ 2 points above baseline (Table 2)
4. Admission order to the hospital
5. Expected length of stay ≥ 48 h (per investigator judgment)

Exclusion criteria

1. Need for renal replacement therapy

Table 1 Index of hospitals currently participating as study recruitment sites

Hospital name and location	Description
Hennepin County Medical Center Minneapolis, MN, USA	Tertiary referral county medical center
MHealth Fairview Ridges Hospital Burnsville, MN, USA	Community hospital
MHealth Fairview Southdale Hospital Edina, MN, USA	Community hospital
MHealth Fairview St. John's Hospital Maplewood, MN, USA	Community hospital
MHealth Fairview University of Minnesota Medical Center Minneapolis, MN, USA	Tertiary academic medical center

2. Need for vasopressors in excess of norepinephrine 0.20 µg/kg/min or the equivalent, as defined by Kotani et al. [26]
3. Need for mechanical ventilatory support in excess of FiO₂ of 60% or PEEP of 5
4. Need for both pressor support at any dose AND positive pressure ventilation of any type (non-invasive or mechanical ventilation)
5. Comfort care only
6. Greater than 48 h since admission
7. Total bilirubin > 3 × or AST/ALT > 4 × ULN
8. eGFR < 25 ml/min/1.73 m²
9. Hemoglobin < 7 g/dL; white blood cell count ≤ 2000/mm³; absolute neutrophil count ≤ 1 × 10⁹/L, or platelet count ≤ 40,000/µL
10. Known HIV, hepatitis B, or hepatitis C
11. Invasive fungal infection (per investigator judgment)
12. Uncontrolled effusions or ascites (per investigator judgment)
13. New/active invasive cancer except non-melanoma skin cancers

14. Known hypersensitivity or allergy to Fisetin.
15. Active treatment with therapies with potential drug-drug interactions (Table 3)
16. Enrolled in another sepsis clinical trial

The initial exclusion criteria for this study excluded patients in ICUs and those on vasopressors or mechanical ventilation with the intent of limiting enrollment to patients with sepsis but not septic shock. The protocol was revised in February 2024 to allow patients in early septic shock, requiring either a modest amount of vasopressor or mechanical ventilation but not both. This change was made to allow broader enrollment, while still limiting participants to those early in their course who would be most likely to benefit from the potential sepsis-abortive effects of the study agent.

Who will take informed consent? {26a}

Patients, or their legally authorized representative (LAR) if the patient is unable to make medical decisions, will be approached for consent by delegated members of the study team with appropriate training and experience in the conduct of human subjects research. At this visit, consents will be reviewed in detail with the subject or LAR. The subject or LAR will have the opportunity to ask questions. If the patient (or LAR) agrees to become a study participant, electronic consent will be obtained using HIPAA and rule 11 compliant electronic consent, stored and maintained in a secure server. For patients who are enrolled through a LAR, we will attempt to re-consent the patient for ongoing study participation. If they decline, previous information gathered will be utilized but no further information will be collected. If the patient is unable to be contacted prior to discharge or by phone following discharge, or they do not regain the

Table 2 Sequential Organ Failure Assessment (SOFA) score

Points assigned	0	1	2	3	4
Cardiovascular	MAP > 70 mmHg	MAP < 70 mmHg no vasopressors	Dopamine ≤ 5 mcg/kg/min	Norepinephrine < 0.1 mcg/kg/min or dopamine > 5 mcg/kg/min	Norepinephrine > 0.1 mcg/kg/min or any dose of vasopressin
Respiratory (PaO ₂ /FiO ₂ ratio)*	> 400	301–400	201–300	101–200	≤ 100
Renal (creatinine, mg/dL)	< 1.2	1.2–1.9	2.0–3.4	4.5–4.9	≥ 5.0
Coagulation (platelet count 10 ⁹ /L)	> 150	100–149	50–99	20–49	< 20
Liver (total bilirubin, mg/dL)	< 1.2	1.2–1.9	2.0–5.9	6.0–11.9	≥ 12.0
Neurologic (Glasgow Coma Scale)	15	13–14	10–12	6–9	< 6

*If PaO₂ is unavailable, SaO₂ / FiO₂ ratio with altered cutoffs are utilized

Table 3 Excluded medications

Alfentanil	Glyburide	Phenobarbital
Amikacin	Idelalisib	Phenytoin
Amiodarone	Imipramine	Plazomycin
Atazanavir	Indinavir	Posaconazole
Atomoxetine	Itraconazole	Rameltoen
Bosentan	Ketoconazole	Rifampin
Carbamazepine	Lithium	Riluzole
Ceritinib	Lomitapide	Saquinavir
Clarithromycin	Lopinavir	Sirolimus
Clozapine	Mefipristone	St John's Wort
Colchicine	Mexiletine	Streptomycin
Conivaptan	Methotrexate	Tacrine
Coumadin (Warfarin)	Midazolam	Tacrolimus
Darunavir	Nefazodone	Telithromycin
Desipramine	Nelfinavir	Theophylline
Diclofenac	Neomycin	Thioridazine
Digoxin	Nitroglycerin	Tizanidine
Duloxetine	Nortriptyline	Trimipramine
Everolimus	Olanzapine	Tobramycin
Flecainide	Ombitasivir-paritaprevir-ritonavir,	Tucatinib
Fosphenytoin	Ombitasivir-paritaprevir-ritonavir	Tyrosine Kinase Inhibitors
Gentamycin	plus dasabuvir	Voriconazole
Glymeperide	Paromomycin	Warfarin

capacity to consent, the participant will continue to participate in study procedures under LAR consent.

Additional consent provisions for collection and use of participant data and biological specimens {26b}

Participants are given the option to consent to being contacted by the study researchers about future research opportunities that may involve collecting more data or biospecimens in addition to those already collected for this study. Additionally, participants are given the option to consent to the collection of a rectal swab for fecal microbiome analysis. Finally, participants may optionally consent to allow genetic testing of their biospecimens. If they consent to genetic testing, they may select a blanket assent to any future genetic testing at the discretion of the researchers, or they may request that they be contacted about future genetic studies and grant consent on a case-by-case basis.

Interventions

Explanation for the choice of comparators {6b}

The intervention drug, fisetin, is provided by our supplier (Sharp Clinical Services, LLC, Allentown, PA, USA) in 100 mg capsules. The placebo, provided by the same supplier, consists of cellulose powder with food dye to match the color and appearance of the fisetin capsules. The

main consideration in choosing this comparator was the similarity of its appearance to fisetin capsules.

Intervention description {11a}

Fisetin is provided in 100 mg capsules. Given the relatively high dose of 20 mg/kg/dose prescribed by our protocol (for a 70-kg individual this would be a 1400 mg dose, resulting in 14 capsules/dose), capsules are opened before administration and the powder is mixed into an emulsion such as apple sauce or pudding and administered orally or by other enteral access over less than 30 min. Subjects randomized to receive two doses of fisetin will receive two such doses 24 h apart. Subjects randomized to receive placebo will receive a matched dose of placebo with capsules similarly opened and mixed with emulsion. In cases where a patient does not complete a dose due to patient-related factors, this will be recorded.

Criteria for discontinuing or modifying allocated interventions {11b}

Study drug dosing may not be administered as planned should a subject choose to willfully withdraw from the study, or if in the opinion of the investigators the subject has developed complications, whether related to the study drug or not, that would make future dosing

dangerous or impossible. This may include the occurrence of a serious safety event believed by the investigators to be at least possibly related to the intervention, worsening of liver function, kidney function or blood cell counts to levels that would exclude the patient from initial participation, or loss of ability to safely administer an enteric medication. Hospital discharge would also result in the discontinuation of future doses of the study drug.

Strategies to improve adherence to interventions {11c}

Subjects will be receiving all doses of study medication while they are inpatients. To help ensure against discharge prior to completion of the study drug, one of the inclusion criteria is that subjects are believed likely to be in the hospital for 48 h. Additionally, to ensure that the study drug is accepted by subjects, capsules are opened and the powder is mixed with an emulsion of the subject's choice, such as apple sauce or pudding.

Relevant concomitant care permitted or prohibited during the trial {11d}

All subjects receive concomitant care for sepsis in line with local standards at the discretion of their treating provider team. Presumably, this includes such interventions as intravenous fluids, antibiotics, and source control measures.

Provisions for post-trial care {30}

Aside from planned study visits at 7 and 28 days to collect data for study outcomes, no specific post-trial cares are planned. In the event of physical injury resulting from the research procedures, medical treatment will be available but not offered free of charge. In addition, financial compensation is not available for wages lost because of injury related to the research protocol. This will be emphasized at the time of consent.

Outcomes {12}

Primary outcome

- Change in the composite cardiovascular, respiratory, and renal sequential organ failure assessment (CRR-SOFA) score at 1 week (± 1 day) from enrollment.

As a phase 2 trial, our sample size is not sufficiently powered to allow mortality-driven outcomes; however, the sequential organ failure assessment (SOFA) score has been established as a valid predictor of outcomes in patients with critical illness including sepsis [27, 28], and change in SOFA score has been demonstrated to

correlate well with mortality in clinical trials [29]. The SOFA score carries challenges, though, for example, a portion of the score relies on an ultimately subjective evaluation of the patient's mental status, and the hepatic portion of the score relies on measurement of serum bilirubin levels, which is not routinely measured in most patients. We therefore plan to use a partial SOFA score including the cardiac, respiratory, and renal scores (CRR-SOFA), as our main indicator of response to therapy. To assess the validity of this score in predicting mortality we performed a retrospective feasibility analysis of 572 elderly patients over a 4-year period who were administered intravenous antibiotics in the emergency department, had a SOFA score ≥ 2 , and were admitted to the hospital floor (ICU and step-down units excluded) of one of the centers participating in the study. That analysis demonstrated a change in CRR-SOFA over 7 days to be at least equivalent to the full SOFA score in predicting mortality, ICU admission, and ICU length of stay (Table 4).

Secondary outcomes

Biological secondary outcomes

- Peripheral CD3+ senescent immune cell quantification at 1 week (± 1 day).
- Quantification of senescent immune cell markers at 1 week (± 1 day). The primary marker of interest is p16^{INK4a}, but others may be measured including gH2AX, SASP factor,
- Hmgb1, LaminB1, and p21.
- Quantification of cytokines using a multi-plex panel aimed to assess those typically secreted from senescent cells in a subset of patients at 1 week (± 1 day).

Clinical secondary outcomes

- Safety of 2 doses of fisetin in patients with sepsis or early septic shock, as determined by differences in adverse events and serious adverse events, in total and by organ system.
- Organ failure-free days at day 28 (28 days minus the last day requiring vasopressors, mechanical ventilation, or dialysis, with death assigned - 1).
- Total SOFA score at day 7.
- Zubrod performance status at days 7 and 28.
- 12-item Short Form health survey (SF-12) score at 7 and 28 days
- All-cause mortality at day 28
- ICU days at day 28

Table 4 Preparatory analysis comparing SOFA and CRR-SOFA to predict outcomes in older sepsis patients

Variable	Standard deviation	Unadjusted OR (95% CI)		β -coefficient
		Mortality	ICU admission	ICU length of stay
Total SOFA (day 7)	1.85	1.36 (1.15-1.60)	1.39 (1.22-1.57)	0.97 (0.35-1.57)
CRR SOFA (day 7)	1.85	1.33 (1.10-1.61)	1.41 (1.22-1.62)	0.98 (0.15-1.82)
Δ SOFA (day 1-7)	1.63	1.39 (1.13-1.69)	1.41 (1.21-1.64)	0.89 (0.35-1.43)
Δ CRR-SOFA (day 1-7)	1.21	1.70 (1.31-2.20)	1.82 (1.47-2.25)	1.24 (0.34-2.14)

Participant timeline {13}

On the day of enrollment, upon completion of informed consent, we will collect initial data sufficient to calculate the subject's SOFA and CRR-SOFA scores, as well as completion of baseline Zubrod and SF-12 surveys, and collection of biospecimens to measure senescent cells and our other biological secondary outcomes. A rectal swab will also be obtained. Study day one will occur either on the same day as enrollment or in the case of enrollments occurring too late in the evening to allow for blood processing, on the following day. At that time the subject will be randomized and receive the first dose of study drug. A second dose of the study drug will follow on study day two. On study day seven, clinical data will again be collected to calculate CRR-SOFA and SOFA scores, the Zubrod and SF-12 assessments will be performed, and biospecimens will again be collected. On day 28 the Zubrod and SF-12 assessments will be repeated and biospecimens again collected if possible. Survival and re-hospitalization will be assessed at days 28 and 90. Adverse event monitoring will occur on an ongoing basis, but specific inquiry about adverse events will occur at least on days 1, 2, 7, 28, and 90 (Table 5).

Sample size {14}

We completed a simulation study under various scenarios to evaluate the type-1 error rate and power of our design. We considered four scenarios; a global null scenario and three scenarios under the alternative hypothesis. The three scenarios under the alternative were considered both with and without futility stopping. We completed 10,000 simulated trials for each scenario. In all cases, Δ CRR-SOFA7 for all three groups were simulated from a normal distribution with a standard deviation of 1. The mean for the control group was set equal to 0 and the mean for the treatment groups was varied across the four scenarios (Table 6). OFFD28 was simulated from a proportional odds logistic regression model with an assumed odds ratio of 2.0 for the association between Δ CRR-SOFA7 and OFFD28. This odds ratio is equivalent to the observed odds ratio for the association between the standardized Δ CRR-SOFA7 (i.e., standardized to have a standard deviation of 1) and OFFD28 in a proportional

odds model fit to our preliminary data. The reference distribution for OFFD28 was set equal to the observed distribution of OFFD28 in our preliminary data. The results of our simulation can be found in Table 6.

Simulation results for the null scenario assume no futility stopping to provide an upper bound on the type-1 error rate. Simulation results for the alternative scenarios are presented both assuming no futility stopping and accounting for the potential for futility stopping to illustrate the impact of our futility stopping rule on power. We see that our design has a family-wise type-1 error rate of 0.024 and power ranging from 0.967 to 0.993 across the three alternative scenarios assuming no futility stopping and ranging from 0.946 to 0.99 after accounting for our futility stopping rule. We also completed an additional simulation under the null hypothesis assuming an odds ratio of 3.0, which is stronger than we observed in our preliminary data to evaluate the impact on our type-1 error rate as this could increase the rate of early termination. Under this scenario, the type-1 error rate was equal to 0.026.

Recruitment {15}

Initially, we recruited patients from four hospitals with a combined size of over 1700 beds. We have since expanded to a fifth hospital with an additional 184 beds (Table 1). Screening occurs broadly across all sites utilizing conditional logic in the electronic medical record to identify patients with easily quantified inclusion criteria (e.g., age ≥ 65). Of those patients identified, medical records are then be hand-screened by study coordinators, and potentially eligible subjects reviewed by a co-investigator to ensure eligibility. Potential enrollees' clinical teams are then notified, and unless they object the patient is approached for consent.

Assignment of interventions: allocation**Sequence generation {16a}**

Our total planned sample size is 220 participants. We will not be using stratification. The first 60 participants will be randomized equally (i.e., in a 1:1:1 ratio) to the control group and two active dose levels. After this run-in period,

Table 5 Table of events

		Study day						
	Enrollment	1	2	3–6	7	8–27	28	90
Eligibility								
Inclusion/exclusion	X							
Informed consent	X							
Randomization								
Randomization		X						
Post-randomization								
Study drug administration		X	X ^a					
Case report forms	X	X	X		X		X	X
Bedside data element for SOFA score (vitals, GCS)	X				X ^b			
Biospecimen sampling	X				X ^b		X ^b	
Rectal swab	X							
Zubrod performance	X				X ^b		X ^b	
SF-12	X				X ^b		X ^b	
SnC measurement	X				X ^b		X ^b	
CRR SOFA calculation	X				X ^b			
Adverse event assessment		X	X	X ^b	X ^b	X ^b	X ^b	X ^b
Survival and rehospitalization assessment							X ^b	X ^b

^a Participants allocated to single-day Fisetin dosing will receive placebo on day 2 to preserve blinding

^b Performed at home via phone or mobile phlebotomy, as relevant, if already discharged

Table 6 Trial simulation illustrations

Scenario	Mean for treatment group 1	Mean for treatment group 2	P (reject dose 1)	P (reject dose 2)	P (reject any dose)
1	0	0	0.014	0.012	0.024
Assuming no futility stopping					
2	0	0.65	0.013	0.979	0.979
3	0.325	0.65	0.265	0.959	0.967
4	0.65	0.65	0.882	0.877	0.993
Accounting for futility stopping					
2	0	0.65	0.013	0.946	0.946
3	0.325	0.65	0.265	0.938	0.947
4	0.65	0.65	0.882	0.974	0.99

the remaining 160 participants will be randomized using Bayesian response adaptive randomization with randomization ratio updates occurring after the 60th participant and for every 20 participants thereafter. For each of these randomizations, the sequence will be generated by a computer with a prespecified seed for replicability.

For the later randomizations, the proportion of subjects allocated to placebo will remain one in three, while the allocation ratios for the two treatment arms will be adapted based on the 7-day change from baseline in the composite cardiovascular, respiratory, and renal sequential organ failure assessment score (Δ CRR-SOFA7) using bayesian response adaptive randomization. This will

occur with the goal of allocating more patients to the arm that appears more efficacious at the time of the most recent randomization. Specifically, the randomization ratio for each of the two active treatment groups (i.e., $j = 1, 2$) will be equal to:

$$0.67 * \left(\frac{P(\mu_j = \max\{\mu_1, \mu_2, \mu_3\} | \vec{y})^c}{\sum_{i=1}^3 P(\mu_i = \max\{\mu_1, \mu_2, \mu_3\} | \vec{y})^c} \right),$$

where μ_t is our current posterior Δ CRR-SOFA7 for treatment group t , $\vec{y}$ represents our extant data for the Δ CRR-SOFA7, and with attenuation parameter $c = 0.8$. This attenuation parameter determines the extent to

which we will change the randomization ratio. If c were 0, we would randomize at a 1:1:1 ratio throughout the course of the trial. If c were 1, the probability of being randomized to the j^{th} active arm would simply be $0.67 * P(\mu_j = \max\{\mu_1, \mu_2\} | \vec{y})$. Our statistical model for $\Delta\text{CRR-SOFA7}$ is given in the “[Statistical methods for primary and secondary outcomes {20a}](#)” section.

The benefit of using Bayesian response adaptive randomization here is that our primary question is as to whether *either* one of the doses is efficacious—we do not care which. With traditional, static randomization schemes, we would either have to power both arms sufficiently to test the efficacy of each arm individually, or to know a priori that one of the arms was likely to be more efficacious and only ensure that the more-promising arm was sufficiently powered. Adaptive allocation allows us both to save on sample size (as compared to the first case) while still “exploring” the responses for the less-promising dose; the fact that some participants are still being allocated to the less promising arm allows us to course-correct if the early difference between arms was due to statistical chance and not to a true difference between the arms.

Concealment mechanism {16b}

Central computerized randomization by web-based interface will be used.

Implementation {16c}

A single unblinded statistician will generate the allocation sequence and subjects will be assigned via a centralized web-based interface. The investigational pharmacy (also unblinded) will fulfill study drug orders based on the allocated group. Enrollment and collection of all study data (SOFA score components, Zubrod, SF-12 assessments, etc.) will be completed by study coordinators and investigators who are blinded to the intervention.

Assignment of interventions: blinding

Who will be blinded {17a}

Matched investigational product and placebo will be provided by the manufacturer to each of the investigational pharmacies. Only the pharmacist and the unblinded statistician responsible for maintaining the randomization module will remain unblinded to treatment allocation. All other members of the study team, including coordinators, investigators, clinicians, and outcome assessors, as well as patients will remain blinded to the study intervention until completion of the study.

Procedure for unblinding if needed {17b}

In the event of a medical emergency, investigators should treat the participant under the assumption that they received an active study intervention. If knowledge of the

participant’s study intervention assignment would impact the choice of treatment, e.g., if a particular treatment would be contraindicated if the participant received an active intervention, an investigator may request local and limited unblinding. The investigator makes this request directly to the site’s study pharmacy team, as these are the only study staff with access to the actual treatment assignment. The number of study staff and clinical staff unblinded to a participant’s study treatment assignment should be kept to a minimum, and these staff will be reminded that the study assignment is still considered blinded for study purposes and must not be casually documented or shared. Any staff who are unblinded should not then be responsible for assessing the relationship of events to study intervention for that participant.

The occurrence of an unblinding event will also be reported to the DSMB and other oversight bodies as required, without disclosing the actual treatment assignment.

Data collection and management

Plans for assessment and collection of outcomes {18a}

Data will be collected on the schedule outlined in the section titled “[Participant Timeline {13}](#)” and in Table 5. Clinical data collection will occur directly from the electronic medical record whenever possible. For patients discharged from the hospital (e.g., on days 7 and 28), clinical data such as vital signs, lab data, and biospecimens will be collected either by patient visit to our research coordinators’ offices or by home visit. Quality of life (SF-12 [30]) and performance status (Zubrod [31]) questionnaires will be conducted either in person at these visits, or by telephone. Data will be collected only by study coordinators or co-investigators who have been specifically trained in the assessment being performed.

Plans to promote participant retention and complete follow-up {18b}

Subjects will have vital signs, subjective data, laboratory tests, and biospecimens collected at study day 7, for many this will be after hospital discharge. Because this requires face-to-face contact, study coordinators will work to schedule this visit at the time most convenient for the subject, and to ensure flexibility for this the protocol allows the visit to be up to one day before or after day seven. Additionally, for those unable to travel to our facilities to have this done, we can provide home visits. At day 28, additional subjective data are collected and additional biospecimens. Again, every effort is made for a face-to-face visit, but if the patient is unable, then the subjective data is collected by telephone and biospecimens are waived. For patients who, after enrolling, decide to limit their participation in the study, every effort will

be made to obtain permission to continue collecting data passively from medical records, and if they are willing, to obtain vital signs and lab work at day 7 to complete the primary outcome. To promote compliance with these visits, subjects are provided a small compensation (\$25) for the completion of each outpatient study visit.

Data management {19}

Data collection is the responsibility of the clinical research staff. Study data will be recorded within a RedCAP (Vanderbilt University, Nashville, TN, USA) database with study personnel being granted access to the database based on the tasks listed in the Delegation of Authority log. Only deidentified data with no PHI will be exported for analysis. The electronic consent (eConsent) forms will be maintained in a University of Minnesota RedCap database with established functionality for this purpose in our group. The database is Health Information Portability and Accountability Act (HIPAA) compliant. A fully de-identified (of all protected health information) database will be generated which will be used for statistical analysis and for monitoring by our DSMB.

Confidentiality {27}

Conversations will be held in a private room whenever possible. Telephone interviews will be performed one on one. The electronic consent process takes place using a secure RedCAP server. All information that identifies study subjects will be handled in accordance with regulatory bodies including HIPAA regulations and the central IRB. This information will be made only available to the principal investigators and study personnel who directly participate in the research calls. Prior to enrollment, participants will sign an authorization to use and disclose protected health information (PHI) for research purposes. All staff will have been trained in the use of PHI. Subjects will be assigned a subject ID number and a link between identifiers will be maintained in REDCap. Only the principal investigators and study nurses will have access to this file. All data will be stored on servers (REDCap and an internal secure data shelter) managed by the University of Minnesota Information Services. No data will be stored on individual research computers, flash drives, etc. These servers, REDCap, and the data shelter meet all data regulatory requirements and are HIPAA compliant.

Plans for collection, laboratory evaluation, and storage of biological specimens for genetic or molecular analysis in this trial/future use {33}

Biologic specimens provided by patients will be transported by courier service with the capacity to transport

biologic materials with due consideration of communicable diseases under Universal Precautions. Blood specimens will be collected for both planned and future analyses. In addition, rectal swabs will be collected for future investigations into the relationship between the gut microbiome, systemic inflammation, and cellular senescence. Specimens will be transported and processed by the University of Minnesota biosafety level-2 translation therapy laboratory and frozen at -80°C until the time of analysis. Samples will be de-identified and labeled only with a patient ID number. Other sites will be encouraged to also obtain these specimens using local resources, with the transfer of collected materials as they are able. Blood processing for biomarkers will follow the laboratory's standard operating procedures.

Statistical methods

Statistical methods for primary and secondary outcomes {20a}

The clinical primary endpoint is the difference in the composite cardiovascular, respiratory, and renal sequential organ failure assessment (CRR-SOFA) score at baseline compared to 7 days from enrollment ($\Delta\text{CRR-SOFA7}$). $\Delta\text{CRR-SOFA7}$ will be analyzed in the Bayesian paradigm and used to guide response-adaptive randomization to the three experimental doses, with interim monitoring guided by the predictive probability of success in a future phase 3 clinical trial of the selected dose level.

Our Bayesian probability model will jointly model $\Delta\text{CRR-SOFA7}$ and organ failure-free days through the first 28 days (OFFD28). The primary analysis at study completion will be based on $\Delta\text{CRR-SOFA7}$ and the joint model with OFFD28 will be used to calculate the predictive probability for interim monitoring.

Let y_i , z_i , and t_i be $\Delta\text{CRR-SOFA7}$, OFFD28, and the randomization treatment assignment for participant i . Our preliminary data indicates that $\Delta\text{CRR-SOFA7}$ is approximately normally distributed, and we model $\Delta\text{CRR-SOFA7}$ as a function of the randomized treatment assigned as follows:

$$y_i|t_i \sim N(\mu_{t_i}, \tau_{t_i}),$$

where μ_{t_i} and τ_{t_i} represent the mean and precision for the treatment group $t_i = 0, 1, 2$. We model $z_i|y_i$ using a proportional odds logistic regression model with common slopes and intercepts across treatment groups. Specifically, let $\gamma_{i,j} = P(z_i \leq j|y_i)$ and,

$$\text{logit}(\gamma_{i,j}) = \alpha_j + \beta * y_i.$$

We place the following priors on all model parameters:

- $\mu_t \sim N(0, \text{precision} = 0.01)$ for $t = 0, 1, 2$

- $\tau_t \sim \text{Gamma}(0.001, 0.001)$ for $t = 0, 1, 2$
- $\beta \sim N(0, \text{precision} = 0.04)$
- $(\epsilon_1, \dots, \epsilon_{29}) \sim \text{Dir}(\xi_1, \dots, \xi_{29})$, where $\xi_i = \frac{2}{29} \forall i$ and
 - $\epsilon_1 = \text{inv.logit}(\alpha_1)$
 - $\epsilon_j = \text{inv.logit}(\alpha_j) - \text{inv.logit}(\alpha_{j-1})$ for $j = 2, \dots, 28$
 - $\epsilon_{29} = 1 - \text{inv.logit}(\alpha_{28})$

The posterior distributions for all model parameters will be approximated via Gibbs Sampling implemented in JAGS using the R package *rjags* [32]. We will be conducting interim analyses at approximately 100 and 180 participants; if there is no early termination, we will declare a dose superior to the control if: $P(\mu_t > \mu_0 | \vec{y}) > \phi_{\text{final}}$ for any dose $i = 1, 2$, where the threshold $\phi_{\text{final}} = 0.984$ was determined via simulation.

The secondary analysis of the clinical primary endpoint will include complementary analyses to augment the primary analysis. $\Delta\text{CRR-SOFA7}$ will be summarized by treatment group using the mean and standard deviation and compared using linear regression with dummy variables for the treatment groups. In addition, we will complete a linear regression analysis that adjusts for age, race, and sex, as well as other covariates that are differentially distributed between treatment groups. Finally, we will analyze the day 7 and day 28 data simultaneously using a linear mixed model with a random effect to account for the correlation between multiple observations from the same individual. We will complete an analysis that adjusts only for baseline, as well as an analysis that also adjusts for age, race, biological sex, SOFA score, and other covariates that are differentially distributed by treatment groups.

The analysis of secondary endpoints will focus on summarizing endpoints by treatment groups and comparing outcomes for each of the two dose levels to the control group and to each other. Continuous endpoints (Total SOFA score, Zubrod performance status, SF-12 score) will be summarized by the mean and standard deviation or median and range and analyzed using linear regression. The primary analysis for these secondary endpoints will include dummy variables for the two treatment groups and adjust for the corresponding baseline measures to improve precision. Model diagnostics for key model assumptions will be completed and we will consider appropriate transformations if model assumptions are violated. Endpoints that are collected longitudinally will also be analyzed using a linear mixed model with a random intercept for participants and the fixed effects described above. Organ failure-free days and ICU days through day 28 will be summarized by counts and proportions and compared between groups using proportional odds logistic regression. Finally, time-to-death will

be summarized by treatment groups using Kaplan-Meier curves and compared using Cox proportional hazards regression with dummy variables for treatment groups. For all secondary endpoints, we will also complete a secondary analysis adjusting for age, race, biological sex, and baseline covariates that are maldistributed between treatment groups.

We will also consider biologic endpoints. Our primary analysis of biologic endpoints will compare the change from baseline to day 7 between patients randomized to each dose of fisetin versus those randomized to placebo. The difference between each fisetin dose and the shared placebo will be summarized by the mean difference; if the change from baseline is skewed, then the outcome will be naturally log-transformed for analysis and the difference summarized by the ratio of the geometric mean. The difference between each dose and the shared placebo will be tested using the two-sample t-test with Dunnett's procedure to account for multiple comparisons with the shared control group. If multiple doses are significantly different from the placebo, pairwise comparisons will be considered. As this is a randomized trial, we anticipate that treatment groups will be balanced for important covariates, and our primary analysis will be unadjusted. In secondary analyses, we will adjust for potential confounders, including age, race, biological sex, baseline SOFA score, and other baseline covariates maldistributed between treatment groups using ANCOVA. In addition, we will analyze the day 7 and day 28 data simultaneously using a linear mixed model with a random effect to account for the correlation between multiple observations from the same individual. We will complete an analysis that adjusts only for baseline, as well as an analysis that also adjusts for age, race, biological sex, SOFA score, and other covariates that are differentially distributed by treatment groups.

Interim analyses {21b}

While the goal of the primary analysis will be to show that the mean $\Delta\text{CRR-SOFA7}$ (defined as CRR-SOFA at baseline— CRR-SOFA at day 7) in each treatment group is lower than in the control group, we also want to consider the predictive probability of success in a future phase 3 trial using OFFD28 as the primary endpoint. Accordingly, interim analysis for superiority and futility will consider both the posterior probability that the mean $\Delta\text{CRR-SOFA7}$ is lower in the treatment group and the predictive probability of success in a future phase 3 clinical trial. In other words, we will only terminate the trial early for superiority if one of the active doses is significantly different than the control and if we are confident that the dose we select would be successful in a future phase 3 clinical trial.

The thresholds for this determination were found by simulation.

Let S be an indicator of success in a future phase 3 clinical trial, where success is defined as detecting a significant difference (i.e., $p\text{-value} < 0.05$) between the selected active dose and the control in a clinical trial with 1000 participants (randomized equally to the treatment and control) with a primary outcome of OFFD28 analyzed using proportional odds logistic regression. In this case, interim monitoring will be based on the predictive probability of success in the future phase 3 clinical trial, which is defined as:

$$P(S|t, \vec{y}, \vec{z})$$

which can be calculated directly from the probability model described in section 20a and approximated via Markov Chain Monte Carlo.

We will complete an interim analysis after enrolling approximately 100 and 180 participants and the trial will terminate for superiority or futility based on the following interim monitoring rules:

- Terminate for superiority if: $P(\mu_t > \mu_0 | \vec{y}) > \phi_{\text{int}}$ and $P(S|t, \vec{y}, \vec{z}) > \psi_s$ for any dose $i = 1, 2$
- Terminate for futility if: $P(S|t, \vec{y}, \vec{z}) < \psi_f$ for all doses $i = 1, 2$

If the trial does not terminate early, we will declare a dose superior to the control if:

- $P(\mu_t > \mu_0 | \vec{y}) > \phi_{\text{final}}$ for any dose $i = 1, 2$

Specific threshold values for declaring superiority and futility were determined by simulation. The final thresholds are:

- $\phi_{\text{int}} = 0.995$ (interim threshold for posterior probability of a lower mean)
- $\phi_{\text{final}} = 0.984$ (final threshold for posterior probability of a lower mean)
- $\psi_s = 0.9$ (interim threshold for probability of success in future trial)
- $\psi_f = 0.075$ (interim futility threshold)

Methods for additional analyses (e.g., subgroup analyses) {20b}

Pre-planned subgroup analysis for all endpoints includes subgroups defined by age, race, biological sex, and SOFA score as a surrogate for severity. Subgroup analyses will follow the same approach as was described above within each subgroup. We will formally test the interaction

between the treatment effect and the subgroups by adding a categorical variable for the subgroup and the subgroup by treatment interaction to the models described above. We will not use formal multiple-comparison adjustments for our subgroup analyses as they are considered exploratory. While the test of the interaction is underpowered, subgroup analyses will provide information about the consistency of the treatment effect across subgroups, which will provide supplementary information for fully understanding of the relationship between the studied interventions and the study endpoints.

Methods in analysis to handle protocol non-adherence and any statistical methods to handle missing data {20c}

The analysis of the primary and secondary endpoints will be completed according to the intent-to-treat principle. Under this principle, all randomized subjects will be included in the analysis in the group to which they were randomized regardless of protocol violations and compliance to treatment assignment. An intent-to-treat analysis requires that complete data are available on all study subjects. Every effort will be made to limit the amount of missing data in this trial. However, some level of missing data is inevitable. In response, we will complete a sensitivity analysis for the primary and secondary endpoints to evaluate the robustness of our conclusions to missing data.

For the sensitivity analysis, we will first compare subjects with and without missing data to identify baseline covariates associated with missing data. In the primary analysis described in Sect. 6.3 of the SAP, which uses a Bayesian probability model to model the association between CRR SOFA and OFFD28, missing data will be handled through data augmentation in the Gibbs sampler. For all other analyses, missing data will be imputed using multiple chained imputation implemented using the *MICE* package [33] in R and combined using Rubin's rule [34]. We will compare the results of these analyses to complete case analyses to evaluate the robustness of our conclusions.

Plans to give access to the full protocol, participant-level data, and statistical code {31c}

There are no plans for public sharing of the full protocol, participant-level data, or biospecimens. Individual requests for access to data or biospecimens will be considered on a case-by-case basis, will comprise only of de-identified data and specimens, and would occur only after completion of a data or materials sharing agreement.

Oversight and monitoring

Composition of the coordinating center and trial steering committee {5d}

Coordination of this trial is occurring from the University of Minnesota Academic Health Center (Minneapolis,

MN, USA). Day-to-day organizational support is managed by a group consisting of the trial PI, individual site PIs, the biostatisticians, and study coordinators from each site. This group routinely meets on a weekly basis to address trial-related concerns or more often as needed.

Composition of the data monitoring committee, its role and reporting structure {21a}

The Data and Safety Monitoring Board (DSMB) will consist of a clinician with experience in sepsis clinical trials, an unblinded biostatistician, and an expert on the science of aging. No member of the DSMB shall be a member of the study staff, have any relation to the sponsor, or be otherwise involved in the trial outside of DSMB activities. The DSMB will meet prior to trial launch. Available safety data will be reviewed with a safety-specific scheduled DSMB meeting after patient 60 (after completion of the 1:1:1 burn-in period). Additionally, safety data will be reviewed during the already scheduled formal interim analyses scheduled after patients 100 (~50%) and 180 (~75%). In addition, if two similar severe AEs or SAEs occur and are judged to possibly be related to the study drug, trial enrollments will be paused until review by the DSMB.

The DSMB may request that enrollment be halted for safety reasons. If a trial is temporarily halted or stopped for safety reasons, IRBs/ethics committees will be informed. The sponsor, in collaboration with the DSMB and relevant health authorities, will determine if it is safe to resume the trial. The sponsor will notify site investigators of this decision. The conditions for resumption of the trial will be defined in this notification. Site investigators will notify their local IRBs/ethics committees.

Adverse event reporting and harms {22}

At each contact with the subject, personnel must seek information as to discomforts or adverse experiences by specific questioning and, as appropriate, by examination. Information about adverse events (AEs) to be recorded includes event description, time of onset, investigator assessment of severity, relationship to Study Agent(s) and Intervention(s), and time of resolution or stabilization of the event. All AEs occurring during the study period will be documented appropriately regardless of relationship to study products or procedures unless included on the list of clinical outcomes that may be exempt from adverse outcomes reporting as part of the study protocol and outcomes. Information on all identified AEs will be recorded within 5 days of recognition.

Serious adverse events (SAEs) are any adverse events that are life-threatening or result in death, require or prolong inpatient hospitalization, result in persistent or significant incapacity, result in a congenital abnormality/birth defect where parental exposure to study drug may

have reasonably impacted pregnancy, or any other medical event that requires intervention to prevent one of the previously listed outcomes. SAEs will be entered into the study database within 24 h of the investigator's awareness of the event.

As patients hospitalized with sepsis are expected to have a high prevalence of a wide range of symptomatology associated with sepsis, most events will be excluded from adverse event reporting *unless* (a) the event is both serious and definitively, probably, or possibly related; (b) the event is unexpected and definitely or probably related; and (c) the event was not preexisting prior to enrollment. Examples of such symptoms include fever, myalgias, arthralgias, fatigue, headache, rhinorrhea, cough, dyspnea, dysuria, rash, lightheadedness, and others. Furthermore, due to the high frequency of laboratory testing, results outside the range of normal are expected as part of the underlying disease process. Abnormal laboratory results will only be reported as adverse events if they (a) were not preexisting prior to enrollment or worsened from values prior to increase, (b) lead to a change in patient management, and (c) they indicate an increased risk to the patient.

Frequency and plans for auditing trial conduct {23}

External auditing will be conducted by the Clinical and Translational Science Institute at the University of Minnesota. Auditors function independently of the investigators and sponsor. Auditing visits will occur at least annually, or more frequently at the discretion of the auditing body.

Plans for communicating important protocol amendments to relevant parties (e.g., trial participants, ethical committees) {25}

Important protocol amendments will be conveyed to the IRB, FDA, and the funding agency (NIA) through official channels established by these agencies for submission and approval of protocol modifications. Investigators will be notified of protocol changes both by provision of the amended protocol and by direct communications by telephone, email, and weekly investigator meetings. The trials registry on clinicaltrials.gov will be updated regularly with changes. We do not anticipate making protocol changes that will affect the subject-facing portions of the trial, but if such changes are made subjects will be informed directly by study staff by telephone.

Dissemination plans {31a}

We anticipate publication of the results of our trial in an established medical journal. We will also update the trial's publicly accessible clinicaltrials.gov entry with aggregate results from the trial.

Discussion

This multi-center, randomized, controlled, double-blinded, adaptive allocation trial will investigate the effects of the senolytic drug fisetin on senescent cell activity and clinical outcomes in patients with sepsis. Our hypothesis is that senescent cells contribute to the maladaptive immune signaling that drives sepsis and septic shock and that decreasing the abundance of these cells will mitigate the clinical deterioration of septic patients and overall improve outcomes. The trial compares fisetin at two different dosages with placebo, and an adaptive allocation protocol will be utilized after the first stage of the trial to modify allocation to favor fisetin at whichever dose demonstrates the greatest efficacy. Our primary outcome for the trial is the change in composite cardiac, respiratory, and renal SOFA scores (CRR-SOFA) between days one and seven. In our preparatory analyses, this outcome correlated well with mortality, need for ICU admission and, for those admitted to the ICU, ICU length of stay. Our intention, should this phase 2 trial show clinical efficacy, is that the findings and experience obtained here will guide a future phase 3 trial.

This study represents a small number of clinical trials of which we are aware that targets the biologic underpinnings of aging. To our knowledge, this is the first trial to specifically target cellular senescence in sepsis, and more generally the only trial of fisetin of which we are aware in the setting of acute infection. The biologic assays nested within the trial will test critical unknowns related to the use of fisetin in human patients. First, it will test the rapidity of clearance of CD3+ senescent cells, as well as the durability at 28 days in the setting of sepsis. Second, it will allow the assessment of a dose-response at the tested dosages. Third, in the setting of sufficient drug-target engagement, the study design will establish whether the drug efficaciously reduces systemic markers of inflammation, the presumed mechanism by which there might be clinical efficacy to reduce organ failure.

From a clinical standpoint, a reduction in mortality is an unreasonable outcome for a phase II trial, which would require several thousand patients for a moderately effective agent. There is no clearly accepted trial outcome for phase II trials in sepsis. An alternative study outcome for sepsis trials being entertained in the field is organ failure-free days, or 28 days minus the last day the patient required either mechanical ventilation or exogenous vasopressors (and in some definitions, renal replacement therapy). The pathophysiologic cascade leading to death is through multisystem organ dysfunction syndrome, the clinical consequence of which for patients is mechanical ventilation, vasopressor support, and renal replacement therapy. These failures take time to manifest, and detecting differences between groups, while requiring

a smaller sample size than mortality, still represents a large cohort. It is worth noting that organ failure can be monitored and quantified using tools such as the SOFA score, consisting of 6 subscores quantifying respiratory, cardiovascular, renal, liver, coagulation, and neurologic function. Due to limitations in certain components of the score (liver, platelet, neurologic), and the fact the first three subscores align with the clinical outcomes of organ support of interest as a potential alternative trial outcome in sepsis, this trial is novel in that it is focused on targeting only the cardiovascular, respiratory, and renal component of the SOFA score (CRR-SOFA). This is the first trial that we are aware of that utilizes this outcome, and the predictive capacity of CRR-SOFA to predict changes in 28 organ failure-free days may represent a novel phase II trial outcome pending the observations of this study.

Trial status

This trial is actively recruiting as of August 23, 2023. The original protocol at launch was version 1.1, dated March 17, 2023. The protocol was updated to the current version (2.0) on February 9, 2024, which included amended exclusion criteria (see section {10} on [Eligibility Criteria](#)) and an additional site (see section {9} on [Study Setting](#)) in order to broaden enrollment. We anticipate completion of enrollment by quarter 3 of 2026.

Abbreviations

AE	Adverse event
ALT	Alanine transaminase
AST	Aspartate transaminase
CRR-SOFA	Composite of the Cardiac, Renal and Respiratory components of the SOFA score
DSMB	Data and Safety Monitoring Board
eGFR	Estimated Glomerular Filtration Rate
FDA	Food and Drug Administration
Fio2	Fraction of inspired oxygen
HIPAA	Health Information Portability and Accountability Act
HIV	Human immunodeficiency virus
ICU	Intensive care unit
IRB	Institutional review board
LAR	Legally authorized representative
MODS	Multi-organ dysfunction syndrome
NIA	National Institute on Aging
PEEP	Positive end-expiratory pressure
PHI	Protected health information
SAE	Serious adverse event
SF-12	12-Item Short Form health survey
SnC	Senescent cell
SOFA	Sequential organ failure assessment

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Authors' contributions {31b}

MS is a biostatistician, developed the electronic data capture instruments, and contributed to protocol development and manuscript preparation. DAW is the site PI for MHealth Fairview sites and contributed to protocol development, patient enrollment, and manuscript preparation. BED is the site PI for Hennepin Healthcare sites and contributed to protocol development and patient

enrollment. AS is a project manager and contributed to protocol development. LJN contributed to protocol development and securing funding. ELS contributed to protocol development related to bioassays. JLK contributed to protocol development and secured the parent IND from the FDA for the use of fisetin. TT contributed to protocol development. TE contributed to protocol development. CS is a biostatistician, developed the electronic data capture instruments (with MS, above), and contributed to protocol development. SV is the unblinded study statistician. JAK is the lead biostatistician and conducted the simulation studies and contributed to protocol development. MAP is the study sponsor, conceived the study, led the proposal and protocol development, and secured funding. All authors read and approved the final manuscript.

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Data availability [29]

Only the study team (investigators, study coordinators, biostatisticians) and the NIA will have access to the final trial dataset. Aside from the PI, the NIA, and the study coordinators who directly collected the data, all other team members will only have access to de-identified data.

Declarations

Ethics approval and consent to participate[24]

The protocol for this trial was centrally approved by the Advarra IRB (Protocol number Pro00070555) and subsequently approved by individual site IRBs (University of Minnesota—SITE00001869 which covers four of the participating hospitals; Hennepin County Medical Center—IRB-FY2023-660 which covers the fifth). Informed consent documented either in writing or by e-consent per local institutional protocols will be obtained from all participants.

Consent for publication [32]

Not applicable. We do not anticipate publication of any details, images, or videos relating to any individual subjects.

Competing interests[28]

MS declares that she has no competing interests. DAW declares that he has no competing interests. BED declares that he has no competing interests. AS declares that she has no competing interests. ELS declares that she has no competing interests. LJN declares that she has no competing interests. TE declares that she has no competing interests. CHS declares that he has no competing interests. JSK declares that he has no competing interests. MAP declares he has no competing interests. SV is funded in part by a Medtronic faculty fellowship and an MSI data science seed grant; he declares that he has no other competing interests. JLK, TT, and Mayo Clinic, where JLK is an emeritus professor, hold patents related to senolytic agents. JLK holds no shares in companies developing senolytics. TT holds shares of Unity Biotechnology. Participation of JLK in this study is in compliance with Mayo Clinic's conflict of interest policies.

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